# ARTIFICIAL INTELLIGENCE – PROJECT 2

## DATA CLASSIFICATION USING AI

### 1. Introduction

Artificial Intelligence enables computers to learn patterns from data and make predictions. In this project, a basic supervised learning classification model is developed using the Iris Flower Dataset.

The objective of this project is to understand the basic classification workflow, including loading a dataset, preparing the data, splitting it into training and testing sets, training a classification model, making predictions, and evaluating the model.

### 2. Project Objective

The main objective of this project is to build a basic classification model using a small dataset.

The project focuses on:

- Loading and understanding a dataset
- Preparing the data for machine learning
- Splitting the data into training and testing sets
- Applying a simple classification algorithm
- Training the AI model
- Making predictions
- Evaluating the model performance

These activities follow the main requirements given for Project 2 in the provided training document.

### 3. Dataset Used

For this project, the **Iris Flower Dataset** was used.

The dataset contains measurements of iris flowers and is used to classify flowers into three different species:

- Setosa
- Versicolor
- Virginica

The dataset contains **150 records and 4 input features**.

### 4. Features Used

The four input features are:

1. Sepal Length
2. Sepal Width
3. Petal Length
4. Petal Width

These measurements are provided to the machine learning model as input data.

## 5. Tools and Technologies

The following tools and technologies were used:

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

## 6. Data Preparation

The Iris dataset was loaded using Scikit-learn.

The dataset was converted into a Pandas DataFrame for easier analysis.

The input variables were stored in **X**, while the target variable was stored in **y**.

The dataset was also checked for missing values.

## 7. Train-Test Split

The dataset was divided into two parts:

- **80% training data – 120 records**
- **20% testing data – 30 records**

The training data was used to train the classification model, while the testing data was used to evaluate its predictions.

## 8. Classification Algorithm

The **K-Nearest Neighbors (KNN)** algorithm was used for classification.

The model was created with:

```
n_neighbors = 3
```

KNN classifies a new data point by considering the nearest data points in the training dataset.

## 9. Model Training

The KNN model was trained using the training data.

The model learned patterns from the flower measurements and their corresponding target classes.

## 10. Prediction

After training, the model was used to predict the classes of the testing data.

The predicted values were compared with the actual target values to evaluate the model.

## 11. Model Evaluation

The model was evaluated using:

- Accuracy Score
- Classification Report
- Confusion Matrix

The accuracy obtained from the Jupyter Notebook should be entered below:

**Model Accuracy: _____ %**

The classification report provides precision, recall, F1-score, and support for each flower class.

## 12. Confusion Matrix

A confusion matrix was created to compare the actual classes with the predicted classes.

The confusion matrix helps identify the number of correctly and incorrectly classified samples for each Iris flower species.

**Insert your Confusion Matrix screenshot here.**

## 13. Results

The trained KNN model successfully classified the Iris flower samples based on their measurements.

The model's performance was evaluated using the testing dataset. The final accuracy should be reported using the actual value obtained from the Jupyter Notebook.

**Final Accuracy: _____ %**

## 14. Conclusion

This project provided practical experience with the basic supervised learning classification process.

The project demonstrated how to:

- Load and understand a dataset
- Prepare input and target data
- Split data into training and testing sets
- Apply a classification algorithm
- Train an AI model
- Make predictions
- Evaluate model performance

Overall, the project helped in understanding the fundamental workflow of **Data Classification Using AI** and basic supervised learning.

## 15. Future Learning

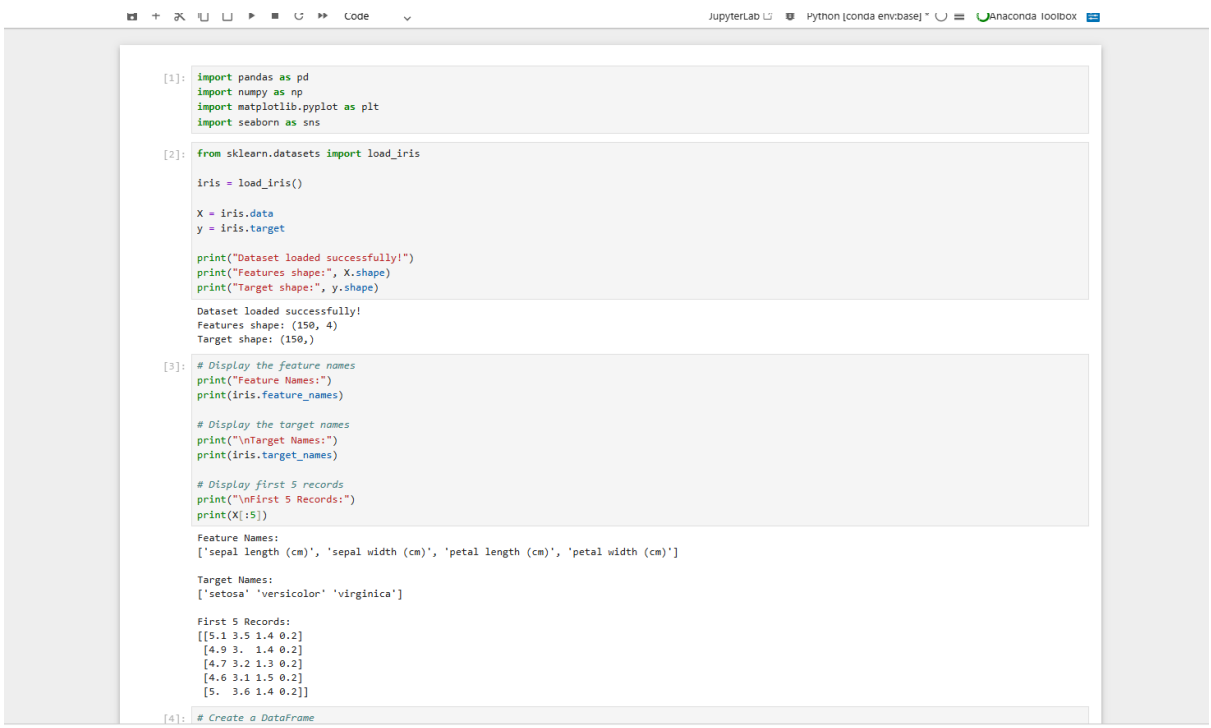
As suggested in the training material, further learning can include experimenting with different algorithms and testing the model with new data.

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# PROJECT DETAILS

**Project:** Artificial Intelligence – Project 2
**Topic:** Data Classification Using AI
**Dataset:** Iris Flower Dataset
**Algorithm:** K-Nearest Neighbors (KNN)
**Training Data:** 120 records
**Testing Data:** 30 records
**Accuracy:** _____ %

**Submitted By:**
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```
[1]: import pandas as pd
     import numpy as np
     import matplotlib.pyplot as plt
     import seaborn as sns

[2]: from sklearn.datasets import load_iris

     iris = load_iris()

     X = iris.data
     y = iris.target

     print("Dataset loaded successfully!")
     print("Features shape:", X.shape)
     print("Target shape:", y.shape)

Dataset loaded successfully!
Features shape: (150, 4)
Target shape: (150,)

[3]: # Display the feature names
     print("Feature Names:")
     print(iris.feature_names)

     # Display the target names
     print("\nTarget Names:")
     print(iris.target_names)

     # Display first 5 records
     print("\nFirst 5 Records:")
     print(X[:5])

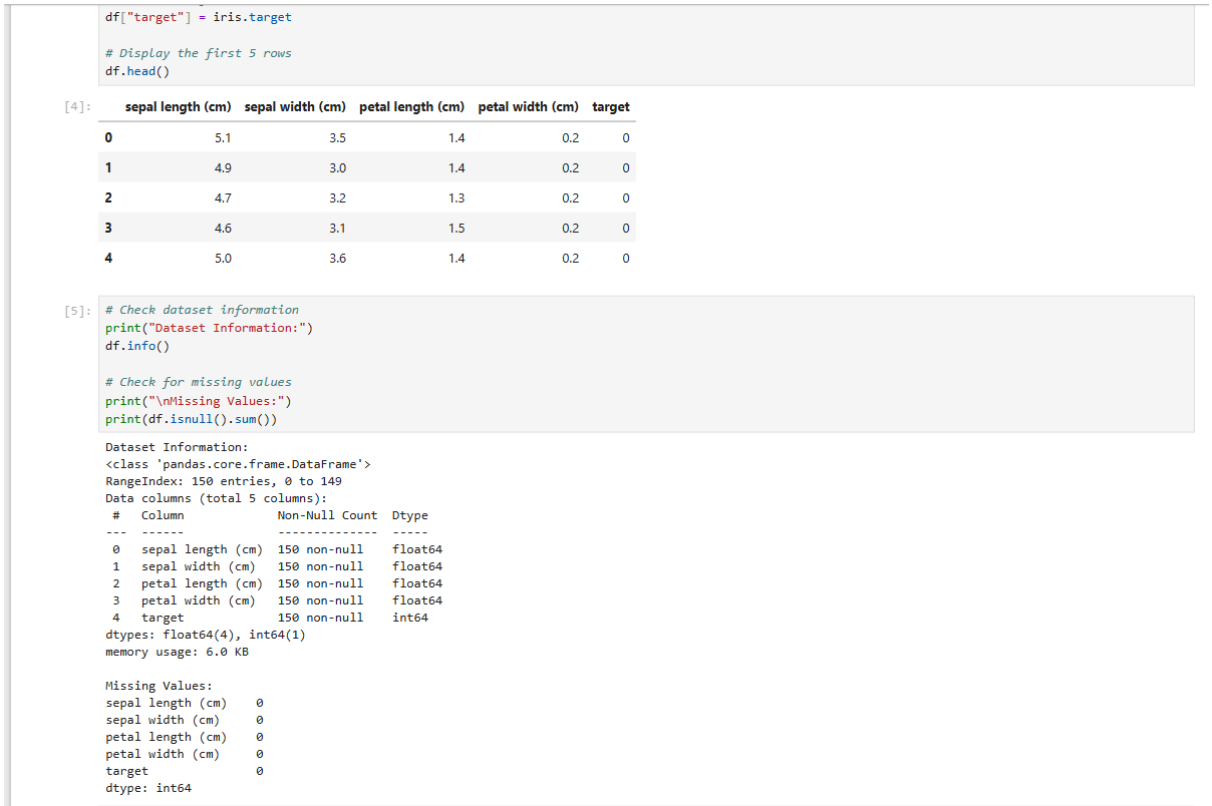
Feature Names:
['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal width (cm)']

Target Names:
['setosa' 'versicolor' 'virginica']

First 5 Records:
[[5.1 3.5 1.4 0.2]
 [4.9 3.  1.4 0.2]
 [4.7 3.2 1.3 0.2]
 [4.6 3.1 1.5 0.2]
 [5.  3.6 1.4 0.2]]

[4]: # Create a DataFrame
```

**Figure 1: Iris Dataset Preview**



```
df["target"] = iris.target

# Display the first 5 rows
df.head()
```

[4]:

| | sepal length (cm) | sepal width (cm) | petal length (cm) | petal width (cm) | target |
|---|---|---|---|---|---|
| **0** | 5.1 | 3.5 | 1.4 | 0.2 | 0 |
| **1** | 4.9 | 3.0 | 1.4 | 0.2 | 0 |
| **2** | 4.7 | 3.2 | 1.3 | 0.2 | 0 |
| **3** | 4.6 | 3.1 | 1.5 | 0.2 | 0 |
| **4** | 5.0 | 3.6 | 1.4 | 0.2 | 0 |

```
[5]: # Check dataset information
     print("Dataset Information:")
     df.info()

     # Check for missing values
     print("\nMissing Values:")
     print(df.isnull().sum())

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   sepal length (cm)  150 non-null    float64
 1   sepal width (cm)   150 non-null    float64
 2   petal length (cm)  150 non-null    float64
 3   petal width (cm)   150 non-null    float64
 4   target             150 non-null    int64
dtypes: float64(4), int64(1)
memory usage: 6.0 KB

Missing Values:
sepal length (cm)    0
sepal width (cm)     0
petal length (cm)    0
petal width (cm)     0
target               0
dtype: int64
```

**Figure 2: Training and Testing Data Split**

Accuracy Percentage: 100.0 %

[13]:
```
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay

# Create confusion matrix
cm = confusion_matrix(y_test, y_pred)

print("Confusion Matrix:")
print(cm)

# Display the confusion matrix
disp = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=iris.target_names
)

disp.plot()
plt.title("Confusion Matrix - Iris Classification")
plt.show()
```

```
Confusion Matrix:
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
```

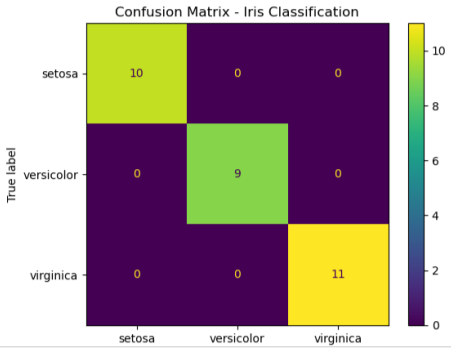
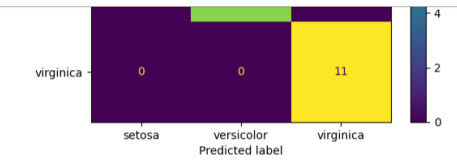



**Figure 3: KNN Model Accuracy**




[17]:
```
comparison = pd.DataFrame({
    "Actual": y_test.values,
    "Predicted": y_pred
})

comparison["Actual Species"] = comparison["Actual"].map(
    {0: "setosa", 1: "versicolor", 2: "virginica"}
)

comparison["Predicted Species"] = comparison["Predicted"].map(
    {0: "setosa", 1: "versicolor", 2: "virginica"}
)

comparison.head(10)
```

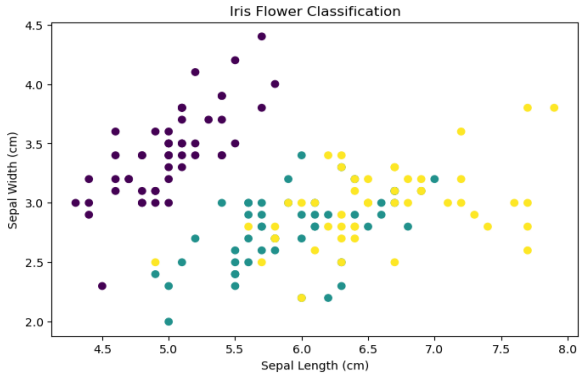
[17]:

| | Actual | Predicted | Actual Species | Predicted Species |
|---|---|---|---|---|
| **0** | 1 | 1 | versicolor | versicolor |
| **1** | 0 | 0 | setosa | setosa |
| **2** | 2 | 2 | virginica | virginica |
| **3** | 1 | 1 | versicolor | versicolor |
| **4** | 1 | 1 | versicolor | versicolor |
| **5** | 0 | 0 | setosa | setosa |
| **6** | 1 | 1 | versicolor | versicolor |
| **7** | 2 | 2 | virginica | virginica |
| **8** | 1 | 1 | versicolor | versicolor |
| **9** | 1 | 1 | versicolor | versicolor |

[18]:
```
plt.figure(figsize=(8, 5))
```

**Figure 4: Classification Report**

```
plt.ylabel("Sepal Width (cm)")
plt.title("Iris Flower Classification")
plt.show()
```




[19]:
```
print("AI PROJECT 2 - DATA CLASSIFICATION")
print("-----------------------------------")
print("Dataset: Iris Flower Dataset")
print("Algorithm: K-Nearest Neighbors (KNN)")
print("Training Samples:", len(X_train))
print("Testing Samples:", len(X_test))
print("Accuracy:", accuracy * 100, "%")
```

```
AI PROJECT 2 - DATA CLASSIFICATION
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Dataset: Iris Flower Dataset
Algorithm: K-Nearest Neighbors (KNN)
Training Samples: 120
Testing Samples: 30
Accuracy: 100.0 %
```

[ ]:

**Figure 5: Confusion Matrix**